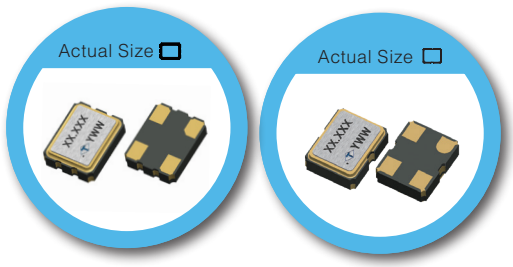
TAITIEN
Our Performance, Your Reputation

OX-B/OY-B Type

High Precision 3.2 X 2.5 / 2.5 X 2.0 mm SMD Crystal Oscillator



RoHS Compliant

FEATURE

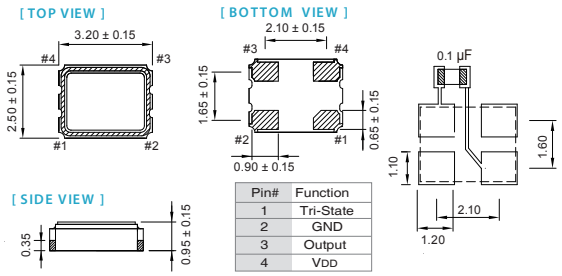
- Tight frequency stability: ±10ppm over -40°C to 85°C ; ±15ppm over -40°C to 95°C
- Operation supply voltage: 1.8V, 2.5V, 3.3V
- LVCOMS Output Logic
- Wide frequency range : 2.5MHz~60MHz

TYPICAL APPLICATION

- Wireless Connectivity
- Mobile device
- Smart meter

DIMENSION & SOLDER PAD LAYOUT (mm)

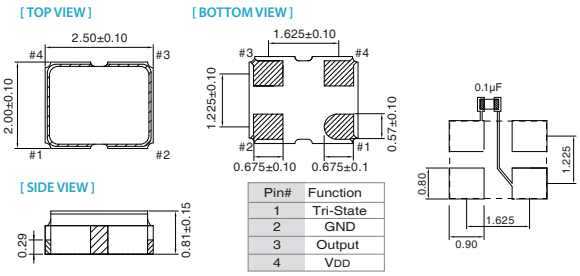
OX-B



Pin#	Function
1	Tri-State
2	GND
3	Output
4	VDD

To ensure optimal oscillator performance, place a by-pass capacitor of 0.1µF as close to the part as possible between Vdd and GND pads.

OY-B



Pin#	Function
1	Tri-State
2	GND
3	Output
4	VDD

To ensure optimal oscillator performance, place a by-pass capacitor of 0.1µF as close to the part as possible between Vdd and GND pads.

ELECTRICAL SPECIFICATION

Parameter		3.3V		2.5V		1.8V		Unit
		Min.	Max.	Min.	Max.	Min.	Max.	
Supply Voltage Variation		V_{DD}-10%	V_{DD}-10%	V_{DD}-10%	V_{DD}-10%	V_{DD}-5%	V_{DD}+5%	V
Frequency Range		2.5	60	2.5	60	2.5	60	MHz
Standard Frequency		24, 25, 32						MHz
Supply Current (@15pf Loading)		-	6	-	5	-	4	mA
Output Level	Output logic	CMOS		CMOS		CMOS		
	Output High	90%V_{DD}	-	90%V_{DD}	-	90%V_{DD}	-	V
	Output Low	-	10%V_{DD}	-	10%V_{DD}	-	10%V_{DD}	V
Transition Time for Standard	Rise Time	-	6	-	6		6	nSec
	Fall Time	-	6	-	6	-	6	nSec
Duty Cycle		45	55	45	55	40	60	%
Start-up Time		-	2	-	2	-	2	mSec
Tri-State	Output Enable	0.7 x V_{DD}	-	0.7 x V_{DD}	-	0.7 x V_{DD}	-	V
	Output Disable	-	0.3 x V_{DD}	-	0.3 x V_{DD}	-	0.3 x V_{DD}	
Stand by Current		-	10	-	10	-	10	uA
Output Loading		-	15	-	15	-	15	pf
RMS Phase Jitter (12kHz to 20MHz)		-	0.5	-	0.5	-	0.5	pSec
Aging (@ 25°C, 1st Year)			±3		±3		±3	ppm
Storage Temp. Range		-55	+125	-55	+125	-55	+125	°C

FREQ. STABILITY vs. TEMP. RANGE

Temp.(°C) \ ppm	±10	±15	±20
-20 ~ +70	○	○	○
-30 ~ +85	○	○	○
-40 ~ +85	○	○	○
-40 ~ +95	×	○	○

*O: Available △: Conditional X: Not available
*Inclusive of calibration @ 25 °C ,operating temperature range,input Voltage variation,load variation.

Note: not all combination of options are available. Other specifications may be available upon request. Specifications subject to change without notice.